

Predicting Post-News Stock Volatility with Machine Learning: Strong Ranking but No Profitable Straddle Strategy Found

1. Introduction

Financial markets react continuously to earnings releases, analyst actions, regulatory filings, product announcements, legal developments, and other company-specific news. Some events produce only a small and temporary response, while others are followed by unusually large price changes and sustained volatility. The goal of this project is to use machine learning to distinguish between those cases.

This project focuses primarily on predicting movement rather than direction. Directional prediction asks whether a stock will rise or fall. Volatility prediction asks how much the stock is likely to move, regardless of whether the movement is positive or negative. This is useful because an event can be market-moving even when its directional effect is difficult to determine in advance.

Volatility prediction also creates a natural connection to option trading. A long straddle purchases both a call and a put at approximately the same strike. The call gains value when the stock rises, while the put gains value when the stock falls. The combined position can therefore benefit from a sufficiently large movement in either direction. If a model can identify which news events will be followed by the greatest movement, a straddle appears to be a reasonable strategy for testing whether that prediction has economic value.

However, predicting high volatility is not the same as identifying a profitable straddle. Option prices already reflect the market's expectation of future movement. A stock can move substantially after an event while still moving less than option prices implied. The trade must also overcome the bid/ask spread, commissions, time decay, changes in implied volatility, and the possibility that a limit order does not fill.

The news feed plays two related roles in the system. First, it identifies the affected company and the time at which a prediction opportunity begins. Second, the headline and structured event information may provide additional information about the event itself. Even a model that uses only market and option features still depends on the news event to determine which stock to evaluate and when to construct its features.

The experiments in this paper follow the full path from prediction to trading. The first experiment evaluates whether a news-triggered machine-learning system can rank which events will be followed by the greatest volatility. The second tests whether explicit event information adds

predictive value beyond the market and option response already observed after the event. The final experiments evaluate whether the volatility signal can be converted into profitable straddle trades using realistic option prices and fill assumptions.

The main result is that the models can rank future post-news volatility, but the straddle strategies tested in this project do not produce robustly profitable executable returns.

2. Data and Experimental Design

2.1 Events, Prediction Time, and Target

Each observation begins with a news event associated with a stock ticker and publication timestamp. For the primary analysis, the prediction is made 20 minutes after the event:

$$d_i = t_i + 20 \text{ minutes},$$

where t_i is the event time and d_i is the decision time.

The model may use only information available by d_i , including the initial stock and volume response, broader market and sector movement, historical option information, macroeconomic conditions, and event information available at that time.

The target is realized variance calculated from five-minute stock returns after the decision:

$$RV_i = \sum_{k=1}^{K_i} r_{i,k}^2$$

where $r_{i,k}$ is the stock return during the k -th observed five-minute bar after the decision time for event i , and K_i is the number of qualifying post-decision bars available before the target period ends.

The target period begins immediately after the 20-minute decision point and continues through 24 hours after the original event. Only observed trading bars are included, and at least 12 post-decision bars are required.

Because realized variance is highly skewed, the regression models predict:

$$y_i = \log \left(\max \left(RV_i, 10^{-10} \right) \right).$$

Here, y_i is the transformed prediction target. The small lower bound of 10^{-10} prevents taking the logarithm of zero and does not represent a financial threshold.

The main prediction task is therefore to use a news event and its initial market response to rank how much movement will occur afterward.

2.2 Data

The study combines company-specific news with stock-market, option-market, macroeconomic, and regulatory-filing information. Benzinga supplied the primary event feed. Five-minute stock, benchmark, and sector data were collected from Alpaca. Historical option-chain information came from MarketData.app, macroeconomic series came from FRED, regulatory filings came from SEC EDGAR, and exact option quotes and trades came from Databento.

Table 1. Main datasets

Dataset	Information used	Event coverage
News events	Ticker, publication time, headline, source, event category, and timestamp quality	538,276 labeled events
Stock and market data	Prior volatility and volume, stock returns, SPY returns, sector returns, and the first 20 minutes of market reaction	400,687 events with early-reaction information
Historical options	Straddle-implied movement, option-price summaries, volume, open interest, and other available chain information	266,724 events with valid pre-event option information
Macroeconomic and SEC data	Market regime, interest rates, economic releases, filing type, filing timing, and disclosure indicators	Available where published before the prediction time
Exact quotes and trades	Bid and ask prices, spreads, fills, executable exits, and option P&L	Exact data collected for 7,485 confirmation events

The option-covered event universe used to form the primary feature-comparison cohort spans January 2016 through June 2026. The final matched volatility cohort contains 134,584 events across 571 tickers. The earliest 35% of these events form the initial training window. The remaining 65% are divided among the five chronological validation blocks, producing 87,480 out-of-fold predictions.

2.3 Data Preparation and Chronological Evaluation

All event timestamps were standardized to UTC. Events without a reliable intraday publication time were excluded from analyses requiring five-minute market alignment.

Duplicate and near-duplicate stories were grouped so that related versions of the same event could not appear in both training and validation. Market features were calculated using only information observed by the prediction time. Historical option features used by the volatility models had to come from before the event, and SEC filings were included only when their official acceptance time preceded the prediction.

The pre-event restriction applies to the historical option features used for volatility prediction. The later trading experiment separately uses contract-level quotes observed at and after the 20-minute decision time to simulate the prices at which a straddle could have been entered and exited.

The models were evaluated chronologically rather than with a random split. The primary analysis used five expanding walk-forward folds. Each model trained on earlier events and predicted a later period. The validation blocks contained sequential groups of events rather than fixed numbers of calendar months. A two-trading-day gap separated training and validation observations, and related duplicate-event groups were removed across each split.

All preprocessing was fitted using only the training portion of each fold. This included numeric imputation, scaling, categorical encoding, and construction of the headline vocabulary.

2.4 Volatility Models and Feature Variants

The broader volatility-modeling stage compared Histogram Gradient Boosting, Random Forest, and Extra Trees regressors. Histogram Gradient Boosting produced the strongest overall volatility forecasts under QLIKE.

The final information comparison used Ridge regression so that numeric, categorical, and headline-text features could be compared with the same estimator. Ridge is a linear regression model that penalizes excessively large coefficients to reduce overfitting. Numeric variables were median-imputed and standardized, while categorical variables were one-hot encoded.

Headlines were represented using term frequency–inverse document frequency (TF-IDF). This representation assigns weights to words according to how informative they are within the training data. The model used both individual words, called unigrams, and two-word phrases, called bigrams. The vocabulary and weights were constructed separately using only the training portion of each fold.

Six information variants were evaluated:

Table 2. Feature variants

Variant	Information included
A: Lagged market	Pre-event volatility, volume, event history, ticker, and time controls
B: Initial market reaction	A plus stock, SPY, sector, and volume response during the first 20 minutes
C: Market, options, and macro	B plus historical option and macroeconomic information

Variant	Information included
D: Structured event information	C plus source, event category, timestamp quality, and SEC context
E: Headline text	C plus headline TF-IDF features
F: Full event information	C plus structured event information and headline text

Variants A, B, and C form a cumulative sequence. Variant B adds the initial market reaction to A, and Variant C adds historical option and macroeconomic information to B.

Variants D, E, and F are separate extensions of C. Variant D adds structured event information, Variant E adds headline text, and Variant F adds both. They should not be interpreted as sequential additions to one another.

This design separates the contribution of the initial market response, option and macroeconomic context, structured event information, and headline text. The incremental comparisons shown in Figure 2 measure each event-information variant relative to the same Variant C baseline.

Different models were used in the later trading experiment because their targets differed from realized-variance prediction. Those model differences are described with the experiment.

2.5 Evaluation Metrics

Spearman rank correlation is the primary volatility metric. It measures whether events assigned larger prediction scores later experience greater realized variance. The main reported value pools all out-of-fold predictions from the five chronological validation folds and calculates one correlation across the combined observations.

QLIKE is used as a secondary volatility-forecasting loss:

$$\text{QLIKE} = \frac{1}{n} \sum_{i=1}^n \left[\frac{RV_i}{\widehat{RV}_i} - \log \left(\frac{RV_i}{\widehat{RV}_i} \right) - 1 \right].$$

where n is the number of evaluated events, RV_i is the observed realized variance, and $\widehat{RV}_i$ is the model's predicted realized variance. Lower QLIKE values indicate better forecasts.

Score-decile analysis is also used to determine whether the largest model scores correspond to a higher concentration of extreme-volatility events.

The incremental event-information comparisons are evaluated on identical observations. Separate 95% cluster-bootstrap confidence intervals are calculated by resampling whole tickers and whole calendar months. Ticker clustering measures sensitivity to the companies represented in the sample, while calendar-month clustering measures sensitivity to the market

periods represented. P-values are adjusted using Holm's method because three event-information comparisons are tested.

The option experiments use executable P&L, fill rate, and premium-weighted return:

$$R_{\text{premium}} = \frac{\sum_i \Pi_i}{\sum_i P_i},$$

where Π_i is the net option P&L for trade i and P_i is the total premium paid to purchase its long option position. Premium-weighted return therefore measures total net P&L relative to the total amount spent on option premiums.

3. Experiments

The experiments address three questions. First, can a news-triggered machine-learning model rank which events will be followed by the greatest remaining volatility after the 20-minute decision point? Second, which information sources contribute most to that ranking? Third, can the predictive signal be converted into profitable long-straddle trades after realistic option prices and execution costs are included?

The first two experiments use 87,480 out-of-fold predictions from the same five chronological validation folds. An out-of-fold prediction means that each event is scored by a model that was not trained on that event. This provides a more realistic estimate of performance on unseen data than evaluating the model on its training observations.

The final experiment uses a smaller, separately collected confirmation cohort with contract-level option quotes and trades. This additional data is necessary because predicting realized variance requires stock prices, while evaluating an executable straddle requires bid and ask prices for the selected contracts at both entry and exit.

3.1 Ranking Future Volatility

The first experiment tests whether larger model scores correspond to larger future realized variance. Spearman correlation summarizes the overall ranking, while the score-decile analysis provides a more intuitive view of what the ranking means.

Within each chronological fold, the validation events were assigned to ten groups from lowest to highest predicted volatility. The boundaries of those groups were calculated using only the training portion of the fold. This prevents information from the validation period from influencing how events are grouped.

For each decile, Figure 1 reports the percentage of events whose future realized variance exceeded the 95th percentile of the training data. This threshold defines an extreme-volatility

event. Because it is a 95th-percentile threshold, approximately 5% of events would exceed it without any useful ranking.

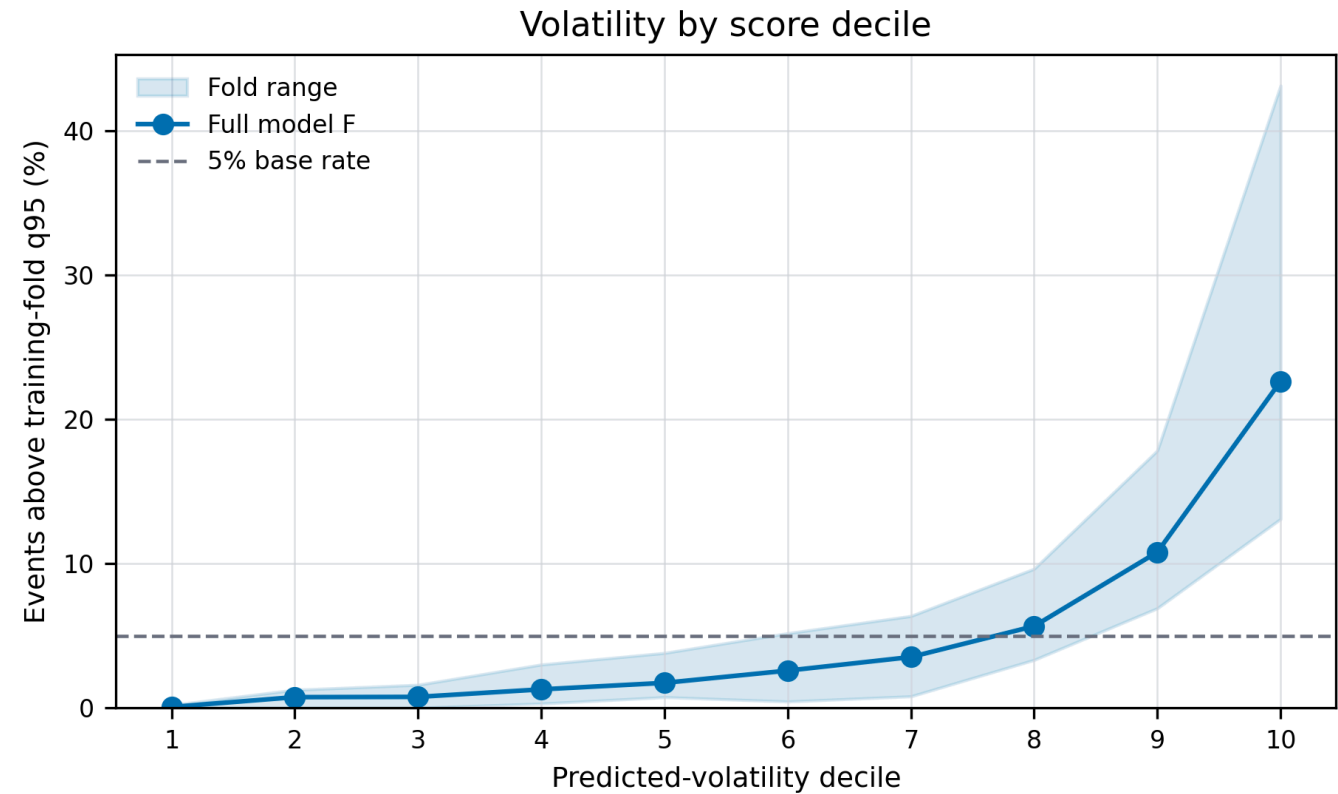


Figure 1. Volatility by score decile. The horizontal axis shows predicted-volatility deciles from lowest to highest. The vertical axis shows the percentage of validation events whose future realized variance exceeds the 95th-percentile threshold calculated from the corresponding training fold. The solid line is the result weighted across five chronological validation folds, the shaded area shows the minimum and maximum fold values, and the dashed line marks the 5% base rate. The shaded area describes variation across folds and is not a confidence interval.

The rate of extreme volatility increases substantially with the model score. Almost none of the events in the lowest-scored deciles exceed the threshold. The rate rises gradually through the middle deciles and then increases more sharply near the top of the ranking. Approximately 11% of events in the ninth decile and 23% of events in the tenth decile experience extreme future volatility.

The highest-ranked decile therefore contains extreme-volatility events at more than four times the 5% base rate. The result is not simply that the top group moves slightly more on average. The model meaningfully concentrates the rare events with the greatest future movement near the top of its ranking.

The wider shaded region in the upper deciles shows that the exact rate varies across chronological periods. Extreme events are rare and can be concentrated in particular market regimes, so some variation is expected. Nevertheless, the highest decile has the greatest

extreme-volatility rate in the combined results and remains well above the base rate across the observed fold range.

For this controlled feature comparison, the full event model achieves a pooled out-of-fold Spearman correlation of 0.626. A positive Spearman correlation means that events assigned larger model scores generally experience larger realized variance. The model also achieves a QLIKE loss of 1.243, compared with 1.868 for the lagged-market variant. Because lower QLIKE is better, this represents an improvement in the quality of the variance forecasts as well as their ranking.

A secondary permutation test provides an additional check on the absolute ranking result. The realized-variance outcomes were randomly reassigned within the validation structure and the correlation was recalculated. The observed full-model performance produced a probability of 0.0001 under this randomized comparison, indicating that a ranking this strong would be very unlikely if the scores contained no information about future volatility.

The first experiment therefore supports the main predictive result: the system can rank which news-triggered events will be followed by the greatest remaining movement after the decision time. The next experiment examines where that ranking ability comes from.

3.2 Sources of the Predictive Signal

The second experiment separates the contribution of different information sources. All six variants use the same Ridge estimator, preprocessing pipeline, chronological folds, and 87,480 observations. Only the feature set changes.

Variants A, B, and C form a cumulative sequence. Variant A uses lagged market information available before the event. Variant B adds the stock, market, sector, and volume response observed during the first 20 minutes after the event. Variant C then adds historical option information and macroeconomic context.

Variants D, E, and F are separate extensions of C rather than additional sequential steps. Variant D adds structured event information, such as the news source, event category, timestamp quality, and available SEC context. Variant E adds the TF-IDF representation of the headline. Variant F adds both structured event information and headline text.

Figure 2 contains two different comparisons. Panel A shows the total ranking performance of each complete feature variant. Panel B isolates the incremental effect of adding event information to Variant C.

Signal sources

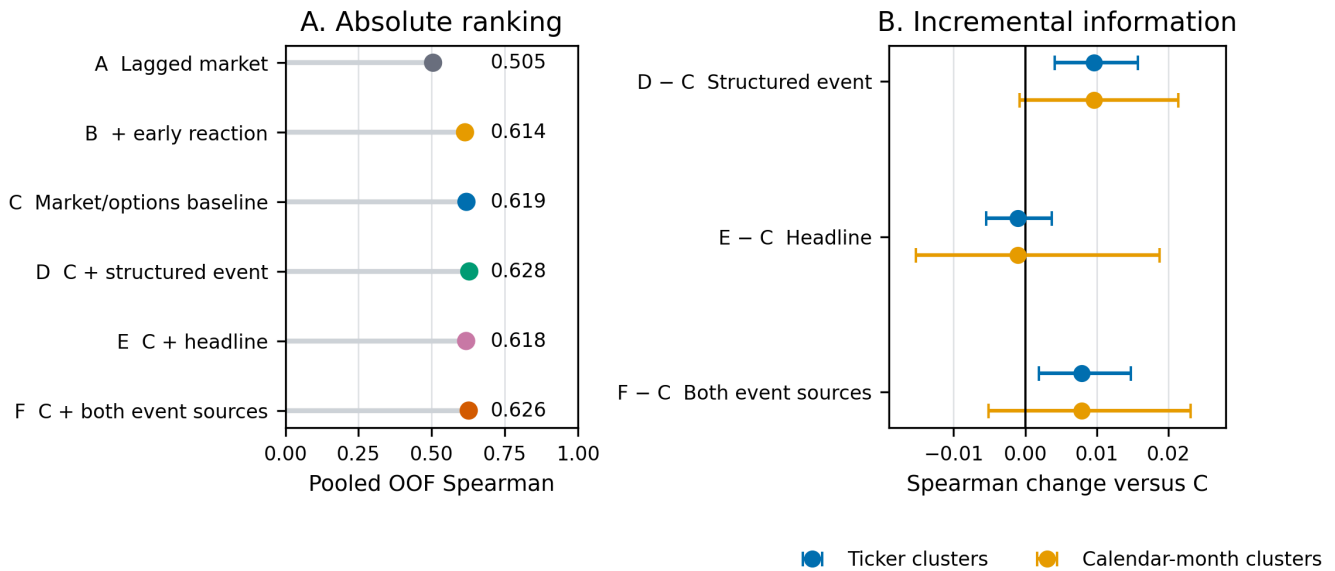


Figure 2. Sources of the volatility-ranking signal. Panel A shows pooled out-of-fold Spearman correlation for feature variants A through F. Variants A, B, and C are cumulative, while D, E, and F are separate extensions of C. Panel B shows the paired change in Spearman relative to C. The blue and orange horizontal lines are separate 95% cluster-bootstrap confidence intervals obtained by resampling whole tickers and whole calendar months, respectively. The vertical line at zero represents no improvement over C.

The lagged-market model achieves a Spearman correlation of 0.505. This indicates that recent volatility, volume, ticker identity, timing, and other pre-event conditions already contain meaningful information about how much movement is likely to occur after a news event.

Adding the initial market reaction increases Spearman from 0.505 to 0.614. This increase of approximately 0.109 is by far the largest improvement in the feature comparison. The early stock and volume response, together with broader market and sector movement, therefore provides most of the additional information used to rank future volatility.

Adding historical option and macroeconomic information increases Spearman more modestly from 0.614 to 0.619. Option prices provide a prior market estimate of expected movement, while macroeconomic features help distinguish between different market regimes.

The three event-information variants remain close to the strong Variant C baseline. Adding structured event information produces the highest absolute Spearman value, 0.628. Headline text alone produces 0.618, slightly below the baseline value of 0.619. Combining structured event information and headline text produces a Spearman correlation of 0.626.

Panel A shows that the complete system ranks volatility well, but these absolute values do not determine whether explicit event information provides a reliable improvement over the strong

market and options baseline. Panel B addresses that narrower question using paired comparisons on the same events.

The structured event variant improves Spearman by 0.00962 relative to C. The full event variant improves it by 0.00784, while headline text alone changes Spearman by -0.00101.

The horizontal lines in Panel B represent uncertainty around these point estimates. Events from the same company may be related because they share liquidity, volatility, investor attention, and other company-specific characteristics. Likewise, events from the same calendar month may be affected by a common market regime.

The ticker-cluster bootstrap therefore resamples entire tickers, keeping all events from a selected company together. The calendar-month bootstrap instead resamples entire months, keeping events from the same market period together. Each method produces a separate 95% confidence interval for the same observed Spearman change. These are alternative views of uncertainty rather than two separate model results.

Under ticker clustering, the confidence intervals for structured event information and the full event model remain above zero. Their positive estimates are therefore not highly sensitive to which companies receive more weight in the sample. Under calendar-month clustering, however, both intervals cross zero. The improvements are less consistent when the represented time periods and market regimes are varied.

Because three event-information comparisons were evaluated, their p-values were adjusted using Holm's method. The adjusted p-values are 0.158 for structured event information, 0.480 for headline text, and 0.251 for the full event model. None satisfy the prespecified threshold of 0.05.

The event-information improvements are therefore small relative to the model's total ranking performance and are not established under the time-clustered, multiple-comparison rule. The strongest result is that most of the predictive improvement comes from market behavior observed around the event. Structured information may contribute a small additional signal, while headline text alone does not improve the ranking.

This result does not mean that the news feed is unnecessary. Every prediction begins with a detected event. The event identifies the company, establishes the timestamp, and determines when the market features should be measured. Variant C is therefore a news-triggered market and options baseline, not a model that continuously scans the market without news.

The comparison instead asks whether the explicit event category and headline add information after the event has been detected and its initial market response has been observed.

3.3 Executable Straddle Returns

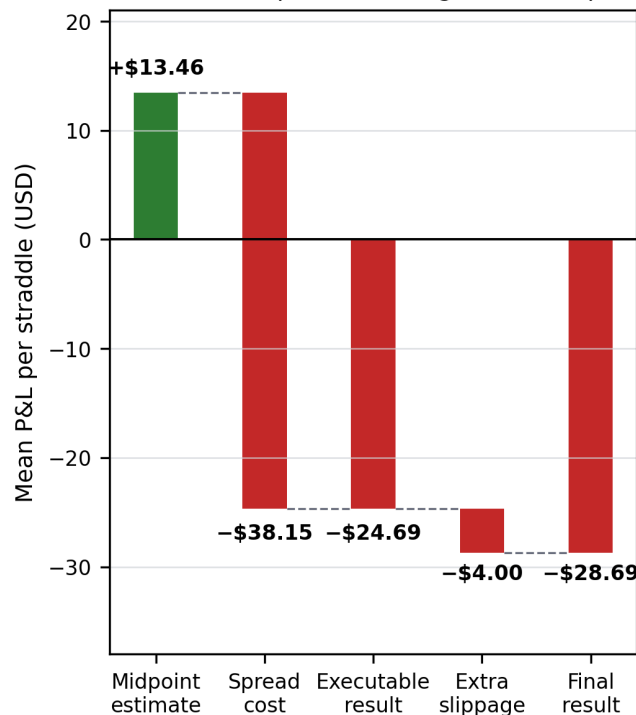
The final experiment tests whether the predicted volatility can be converted into profitable long-straddle trades. Two trading policies were specified before the confirmation results were examined, so their models and selection thresholds could not be adjusted after seeing their performance.

H2 is a logistic-regression classifier that estimates the probability that a straddle will produce positive net P&L. H3 is a multi-output Ridge regression model that predicts several continuous outcomes related to volatility, stock movement, and option returns. H3 uses its predicted return on premium to rank and select trades. Both policies use market and option information available at the 20-minute decision time, but H2 directly predicts profitable versus unprofitable trades, while H3 attempts to estimate the magnitude of several related outcomes.

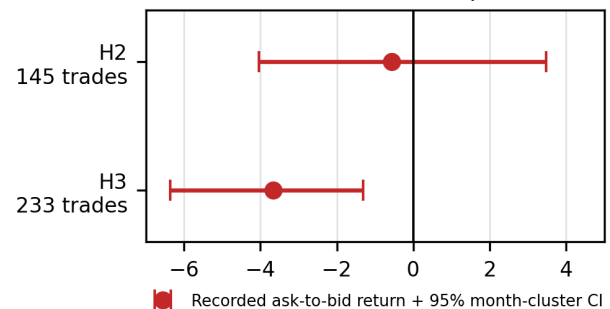
Both policies purchase a near-at-the-money call and put with five to ten calendar days remaining until expiration. Historical contract-level quotes are used to simulate execution. Each option is purchased at the first eligible ask price after the decision time and sold at the first eligible bid price at the same clock time during the following trading session. This includes the cost of crossing the bid/ask spread rather than assuming that every trade occurs at the more favorable midpoint.

Prediction-to-trading gap

A. Execution-cost example across eligible development events



B. Predefined straddle policies



C. Slightly worse execution

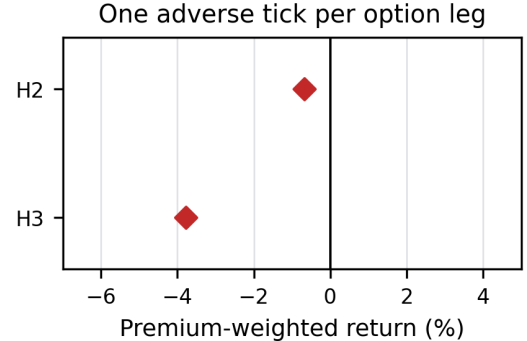


Figure 3. Prediction-to-trading gap. Panel A shows how the mean P&L of a simple reference strategy applied to all eligible development events changes as realistic execution costs are added. Panel B shows the premium-weighted returns of H2 and H3 calculated from recorded contract-level quotes, with calendar-month-clustered 95% confidence intervals. Panel C tests

how the same trades perform if their simulated execution prices are one minimum price increment worse than the recorded quotes, representing a small amount of possible execution slippage. Panel A uses a development cohort and is separate from the H2 and H3 confirmation results.

Panel A demonstrates the difference between midpoint valuation and executable prices. It averages the results across eligible development events with complete entry and exit quotes; it does not represent a single trade or either trading model. At midpoint prices, the reference straddles have an estimated mean P&L of positive \$13.46 per straddle. The combined bid/ask spread creates an average cost of \$38.15, changing the result to negative \$24.69. Assuming execution is one additional minimum price increment worse on each option leg reduces the mean to negative \$28.69.

Panel B reports the confirmation results for the two model-selected policies. Premium-weighted return is total net P&L divided by the total premium paid across the selected trades. The vertical line at zero separates profitable and unprofitable returns.

H2 selects 145 trades and produces a premium-weighted return of -0.558%. Its calendar-month-clustered 95% confidence interval ranges from -4.037% to 3.485%. Because the interval crosses zero, the confirmation data do not establish that H2's return differs from zero.

H3 selects 233 trades and produces a return of -3.661%. Its calendar-month-clustered 95% confidence interval ranges from -6.365% to -1.314%. Because the entire interval is below zero, the confirmation evidence indicates that this policy performed negatively.

Panel C asks whether the conclusion changes if real-world execution is slightly worse than the displayed prices used in Panel B. Option prices move in minimum increments, called ticks, and a displayed quote can change before an order is executed. The sensitivity therefore shifts the simulated prices one tick against the strategy for each option leg and recalculates the returns on the same H2 and H3 trades. H2 decreases from -0.558% to -0.670%, while H3 decreases from -3.661% to -3.777%. This is not a separate model or an assumption that every trade would necessarily receive the worse price; it is a conservative stress test. Both policies are already negative under the main ask-to-bid assumption, and a small amount of additional slippage makes their returns slightly worse.

Neither policy therefore produced robustly profitable executable straddle returns.

3.4 Explaining the Prediction-to-Trading Gap

The volatility and trading experiments are not contradictory because they measure different outcomes.

The volatility model is successful when events with larger scores experience more future movement than events with smaller scores. Spearman correlation depends on the ordering of

realized variance and does not consider the price of entering an option position.

A straddle has a more demanding requirement. The call and put premiums already reflect how much movement option traders expect. The position is profitable only if the realized movement and resulting option values exceed the amount paid for both options, the bid/ask spread, and other changes in the contracts over the holding period.

An event can therefore be correctly ranked as highly volatile and still produce a losing straddle. For example, a stock may move much more than most other stocks in the sample, but the option market may have charged an even larger premium because it also expected an unusual move.

The 20-minute decision period contributes to this distinction. Figure 2 shows that the observed market reaction during those 20 minutes provides most of the improvement in volatility ranking. That reaction helps the model recognize an important event, but it also gives stock and option traders time to update prices. By the time the model is ready to enter, part of the useful information may already be reflected in the straddle premium.

Execution costs create another direct obstacle. Panel A of Figure 3 shows that the quoted spread is large enough to reverse a positive midpoint estimate. This spread effect is directly observed in the exact quote data. Other mechanisms, such as changes in implied volatility and time value, may also contribute, although the experiments do not isolate each mechanism separately.

The combined result is therefore specific. The models identify which news-triggered events are likely to have greater remaining volatility, but the tested policies do not identify volatility that is underpriced by enough to support profitable executable straddles.

4. Conclusion

This project evaluated whether machine learning could rank the volatility that follows company-specific news and whether those predictions could support profitable long-straddle trades. The full event model achieved an out-of-fold Spearman correlation of 0.626, and approximately 23% of events in the highest predicted-volatility decile exceeded the extreme-volatility threshold, compared with a 5% base rate.

Most of the predictive improvement came from the stock, market, sector, and volume response observed during the first 20 minutes after the event. Historical option and macroeconomic information added a smaller improvement. Structured event information produced a small positive effect, but it was not stable across calendar months, while headline text alone did not improve the ranking over the strong market and options baseline. News remained essential for identifying the affected company and prediction time, but much of its useful information appeared to be reflected quickly in market behavior.

The predictive signal did not translate into profitable straddles. A development reference that appeared to earn \$13.46 per straddle at midpoint prices lost \$24.69 after the bid/ask spread was included. The two predefined trading policies produced premium-weighted confirmation returns of -0.558% and -3.661%. These results show that correctly predicting large movement is not enough: the movement must also exceed what the option market has already priced, along with the cost of execution.

The main conclusion is therefore that post-news volatility is predictable, but the tested strategies did not identify volatility that was sufficiently underpriced to produce profitable executable straddles. Future work should target movement relative to option-implied expectations or contract-level P&L directly, potentially using earlier decision times and a larger history of contract-level option quotes and trades.